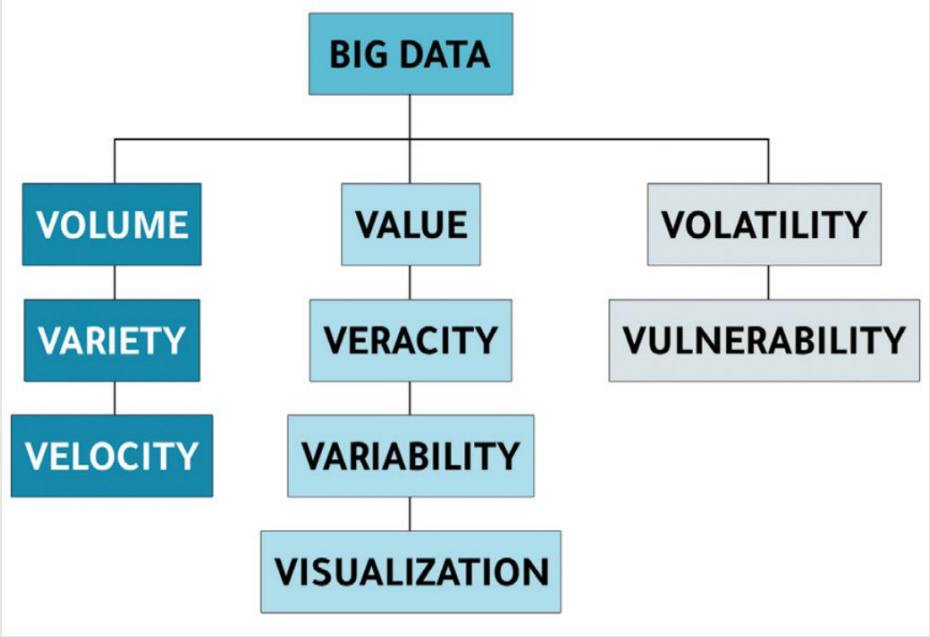
BIG DATA
HOUSING

Big Data 101

- **Big Data** can be defined as data or information characterized by a **Volume**, **Velocity**, and **Variety** so significant/high that they require specific technologies and analysis methods to transform them into value



Data by data, Big Data. Hackenberger, 2019.

- Other "V's" :
  - **Variability**
  - **Veracity**
  - **Visualisation**
  - **Value**

- 2.5 Trillions of Data Bytes created each days

The project

- Develop an innovative app for comprehensive information and analysis of the housing market in France.
- Offer a user-friendly and interactive experience for exploring housing statistics and insights.
- Your application should be accessible to both individuals and professionals

BIG DATA HOUSING

Statistical, textual, and cartographic analysis of real estate in France.

Project's Vs :

+ Volume
+ Variety
+ Volatility

Data variety

Different data formats

- A type of format correspond to a type of information
- Diversifying formats to vary the angle of analysis

Origines variées

- Attaining a better represantation of information
- Depending on the origin, concernes are not the same

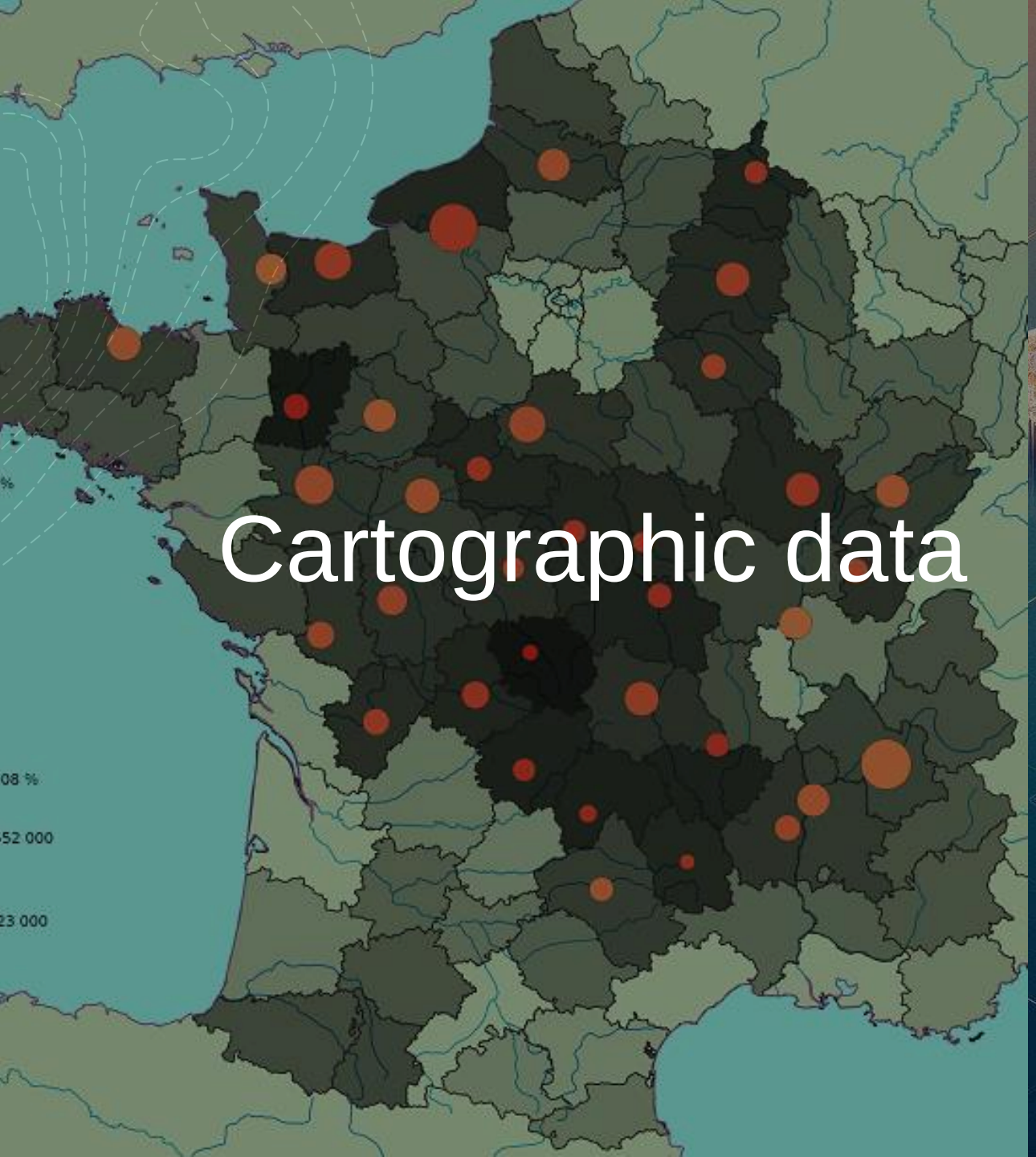
Diversity of analysis levels

- Having a global but also local pespective on the topic of interest

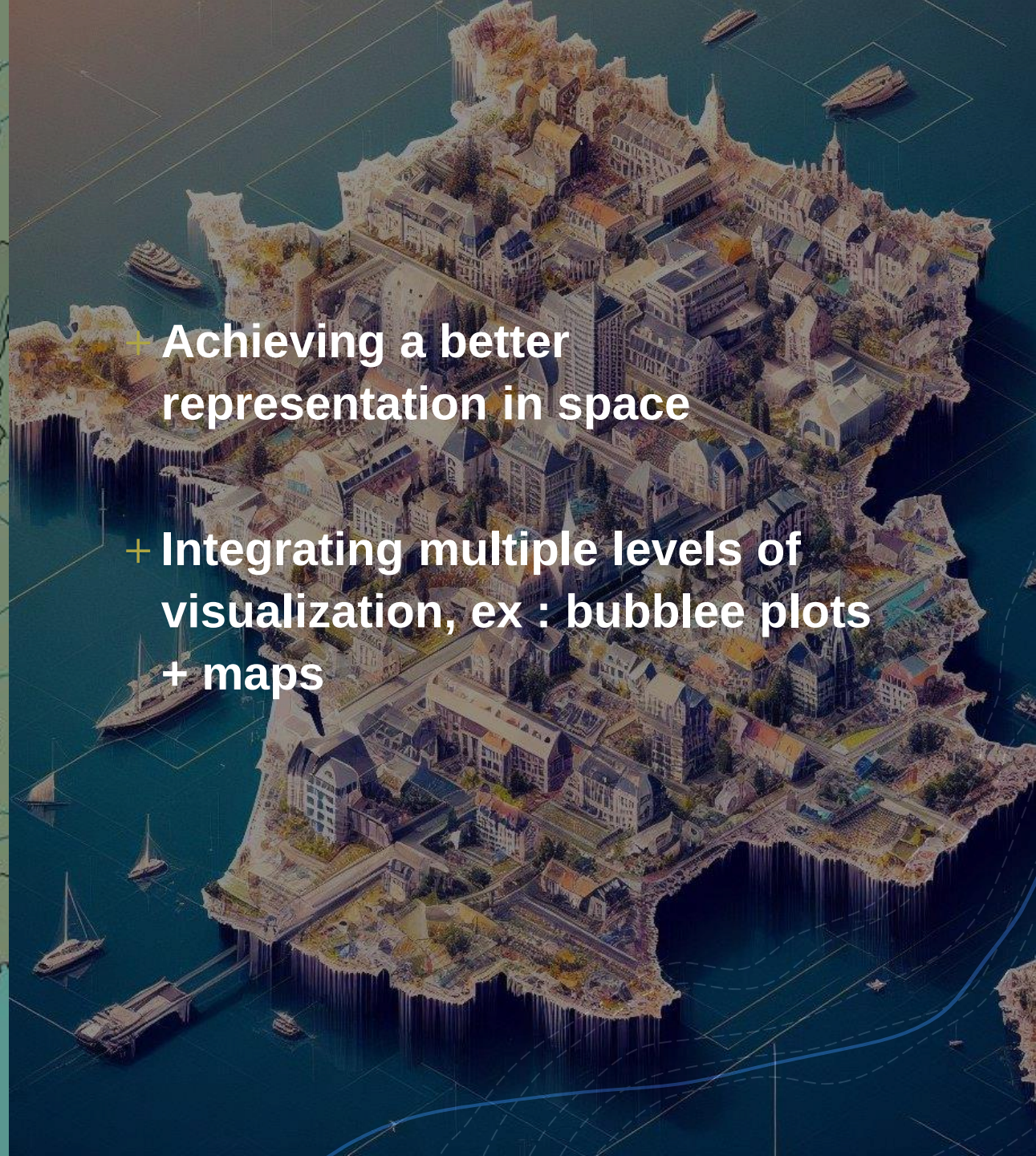
Non relational Databases

- The use of an appropriate database helps optimize storage

- Due to the wide variaty/volatility of data it's advantageous to have a suitable storage system



Cartographic data



- Achieving a better representation in space
- Integrating multiple levels of visualization, ex : bubblee plots + maps

The Data

Gouvernemental datasets

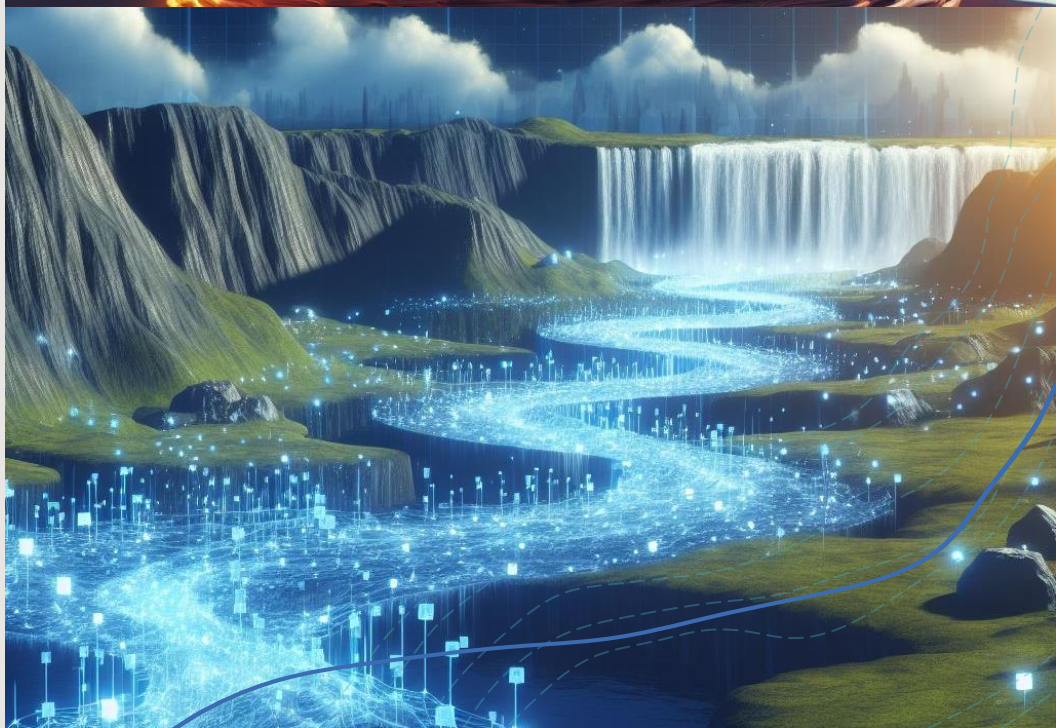
+ Using verified data
+ Open-source disponibility
+ API's

Other Datasets

+ Specific data
+ Different formats of data from DataGouv
+ Attention to usage conditions for scraping data

Processing

- Using appropriate tools to process large volumes of data
- Helps shorten calculation times
- Managing calculations in the cloud

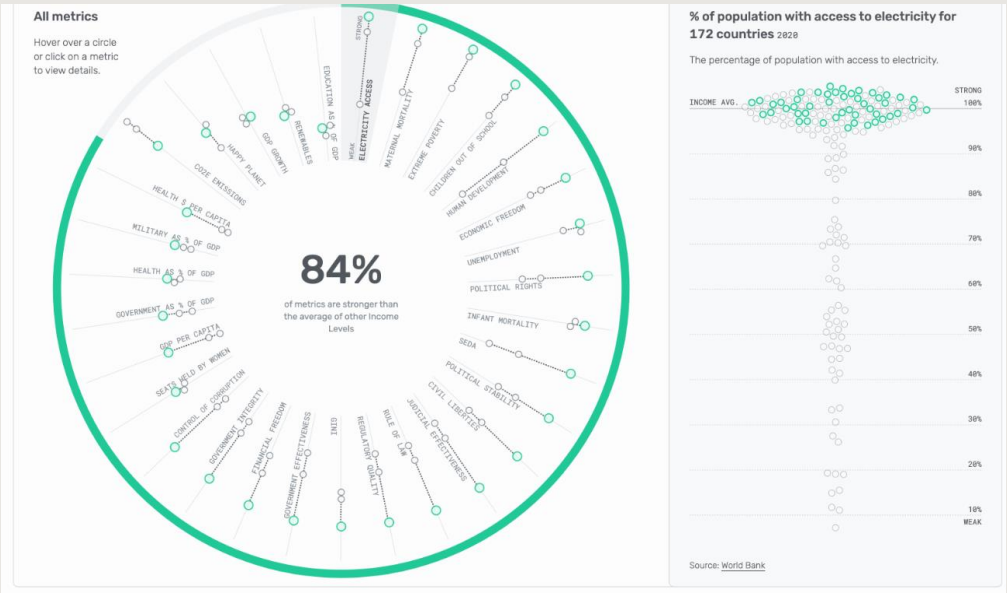


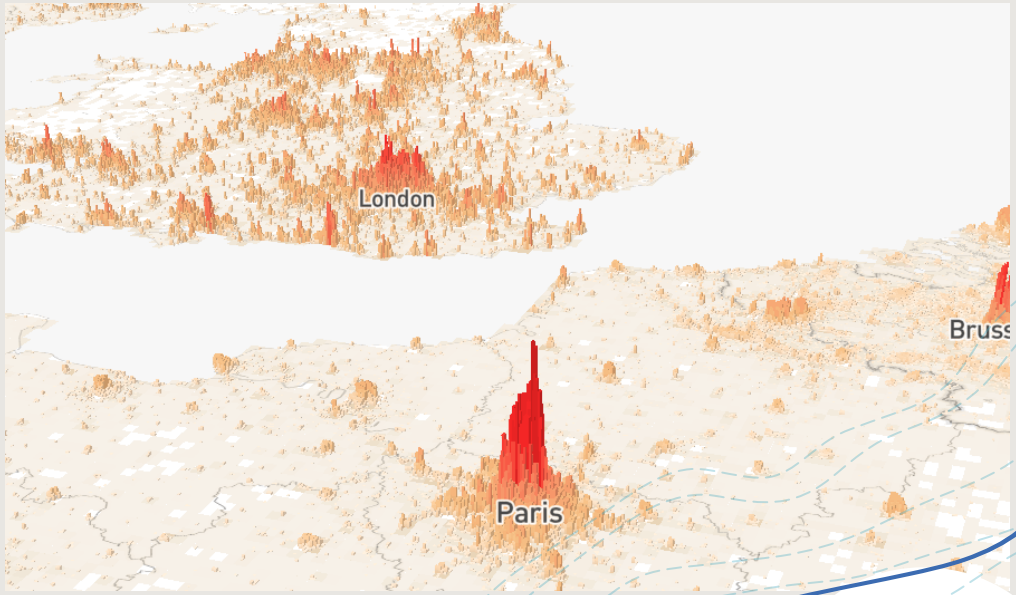
Visualizing data, what for ?

- Visualizing data is making sense of information
- Representing data in high dimensions : AI ?

Dynamic visualization

- Making data interactive :
  - Users can select the information that interests them
  - Filtering information to make the dashboard more readable

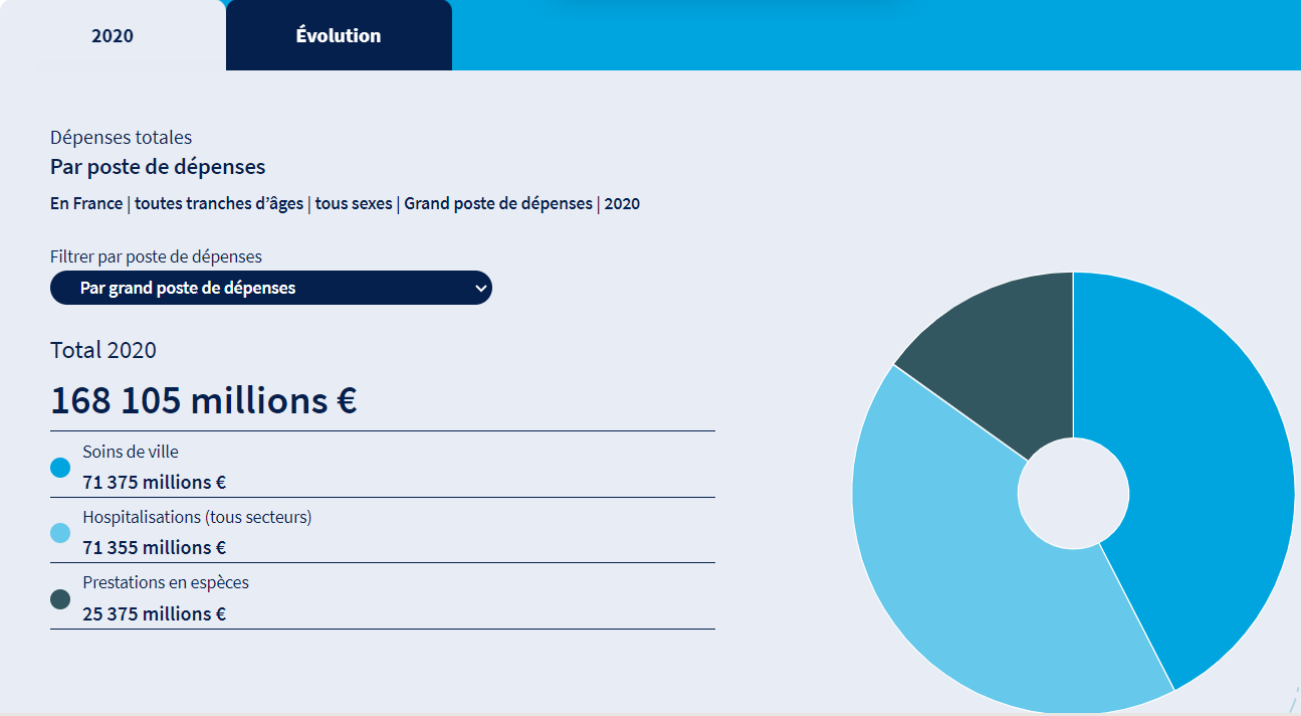
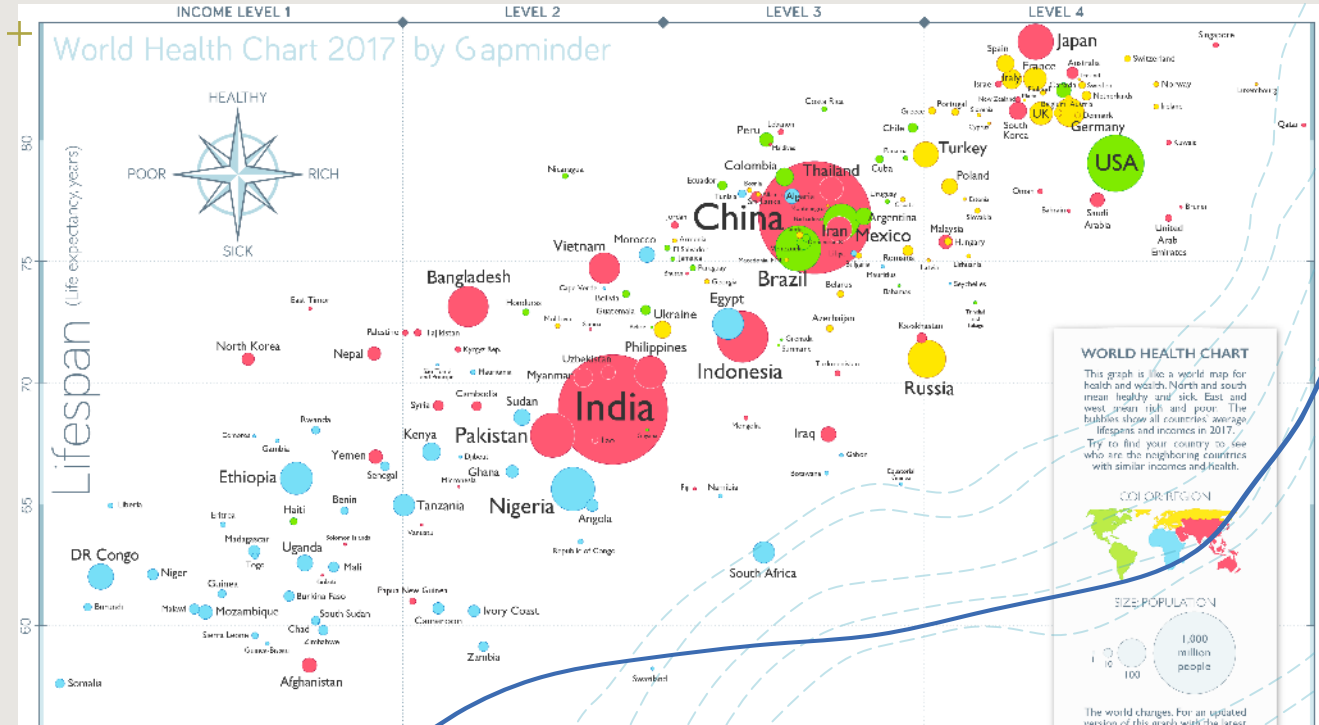




Real-Life examples for inspiration

+ Gapminder : it promotes a fact-based worldview through data visualization.

+ https://www.villesavivre.fr/ : a website for reviews and comparisons of communes in France.

+ Data pathologies : an initiative from Ameli, the french social care system.

+ Numbeo: provides information on various aspects of living, including housing costs, quality of life, etc.